Automotive Radar Array Antenna @ 77GHz

Full-wave design of an automotive radar antenna array on RT/duroid 5880 – from the Hammerstad first cut through impedance re-tuning to beam steering

Simulation report
Tool: openEMS (FDTD) through the VisualopenEMS project layer
Frequency band: 74–80 GHz, centre 77 GHz
2026-11-10
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1 Introduction

This study documents the complete design of a planar 4×4 patch antenna array for automotive 77 GHz radar. The starting point is a single probe-fed rectangular patch on RT/duroid 5880. The element is dimensioned with the classical Hammerstad approximation, characterised in the full-wave model and then matched to 50 Ω with a systematic re-tuning procedure. The resulting element is assembled into a 4×4 array with λ0/2 spacing and studied in two modes of operation: in phase (broadside) and with a progressive phase for a beam steered to +30°.

1.1 Scope

- Single element on RT/duroid 5880 at 77 GHz, matched to 50 Ω by a probe inset.
- 4×4 array with λ0/2 spacing (d = 1.947 mm) and 16 independent 50 Ω ports.
- Broadside operation and phased operation (+30°) in one complete FDTD run each.
- Assessment of directivity, efficiency, coupling and beam steering; comparison with the array factor.

1.2 Results at a glance

The final element reaches an input impedance of Zin = 53.8 + 14.5j Ω and a reflection of -16.9 dB at 77 GHz with a directivity of 9.38 dBi. The broadside array reaches 17.91 dBi directivity; after correcting the power budget the gain is 17.69 dBi at an efficiency of 0.949. The steered variant reaches 17.75 dBi towards θ = 28°, φ = 0°.

2 Model, tools and discretisation

2.1 Simulation chain

The whole study is script driven and reproducible without a graphical user interface. The geometry exists in exactly one place, the Python module `array_model.py`; all project files, runs and evaluations are derived from it.

1. `array_model.py` – substrate, patch, grid, ports, mesh, NF2FF box and the four project variants.
2. `make_project.py` – writes the `.fcsim.json` projects and the construction figures.
3. `validate_project.py` – checks collisions, mesh rules, port positions and the NF2FF box before the run.
4. `run_headless.py` – generates the openEMS script from the project and runs FDTD including post-processing.
5. `plot_results.py` – reads S-parameters and far fields, produces the result figures and `docs/results.json`.
6. `build_docs.py` – builds this report in German and English.

2.2 Discretisation

The mesh is created with the VisualopenEMS preset “high” and supplemented with explicit cell sizes: maximum cell size 0.25 mm, refined edge 0.10 mm, minimum cell size 0.07 mm and minimum z cell size 0.06 mm. The 0.254 mm laminate is therefore resolved by four cells in z and the metal edges by about three cells. The computational domain is enclosed by a PML on all sides; the air clearance is 4 mm to the sides and 6 mm above the board.

The run stops at −40 dB residual energy (openEMS criterion 1e-4), which all four production runs reach after about 10,600 time steps. The excitation is a Gaussian pulse with f0 = 77 GHz and fc = 3.0 GHz, which safely covers the 74–80 GHz band.

3 Antenna construction

3.1 Substrate and stack-up

RT/duroid 5880 with εr = 2.2 and tan δ = 0.0009 in the standard 0.254 mm (10 mil) thickness serves as the carrier. Both copper layers are modelled as 35 µm thick PEC sheets. Figure 1 shows the cut through one patch row; the z axis is exaggerated by a factor of six so that the 0.254 mm laminate remains visible.

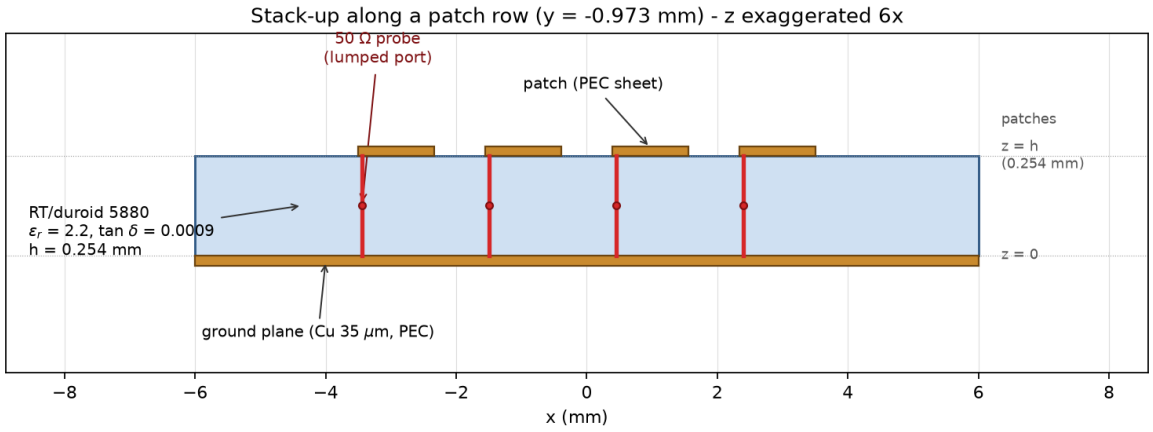


Figure 1. Stack-up along one patch row (y = −0.973 mm). The probes connect the ground plane to the patches; z is exaggerated 6×.

Table 1. Substrate and stack-up parameters.

Parameter	Value
Substrate	RT/duroid 5880
Relative permittivity εr	2.2
Loss tangent tan δ	0.0009
Substrate height h	0.254 mm
Copper thickness (PEC)	35 µm

Parameter	Value
Board size	12.0 × 12.0 mm
Free-space wavelength λ0(77 GHz)	3.893 mm

3.2 Patch dimensioning

The resonant length follows the Hammerstad approximation for a rectangular patch on an infinitely extended substrate. With W = 1.539 mm the effective permittivity is 1.9475 and the starting length is L = 1.1383 mm. The width is chosen so that the radiation resistance is a few hundred ohms and the probe inset covers the required transformation range.

```
eps_eff = (eps_r + 1)/2 + (eps_r - 1)/2 * (1 + 12*h/W)^(-0.5)
dL = 0.412*h * (eps_eff + 0.3)*(W/h + 0.264) /
 ((eps_eff - 0.258)*(W/h + 0.8))
L = c0/(2*f0*sqrt(eps_eff)) - 2*dL
```

Figure 2 shows the dimensioned element. The zoom on the right reveals the feed: in the final design the inset is only 0.058 mm, about 5 % of the patch length – a value that is no longer resolvable in the overall view.

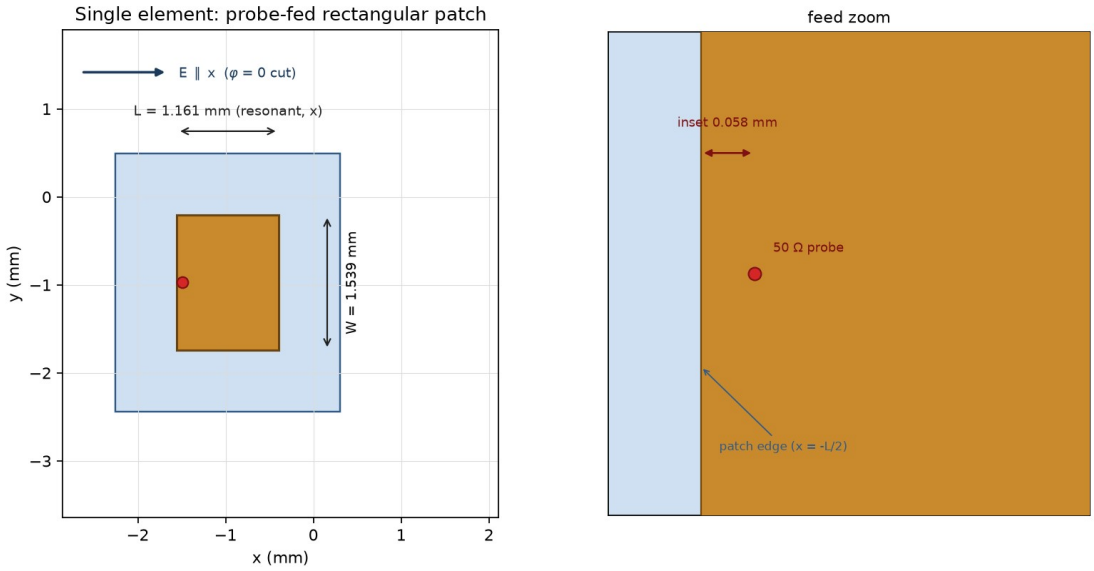


Figure 2. Dimensioned single element (left) and zoom on the probe feed (right).

3.3 Array layout

Four elements in x and four in y form a square array with d = λ0/2 = 1.947 mm spacing. The total footprint of the patch area is 7.00 × 7.38 mm on a 12 × 12 mm board. Each element has its own coaxial probe port from the ground plane to the patch underside, so all 16 elements can be driven independently.

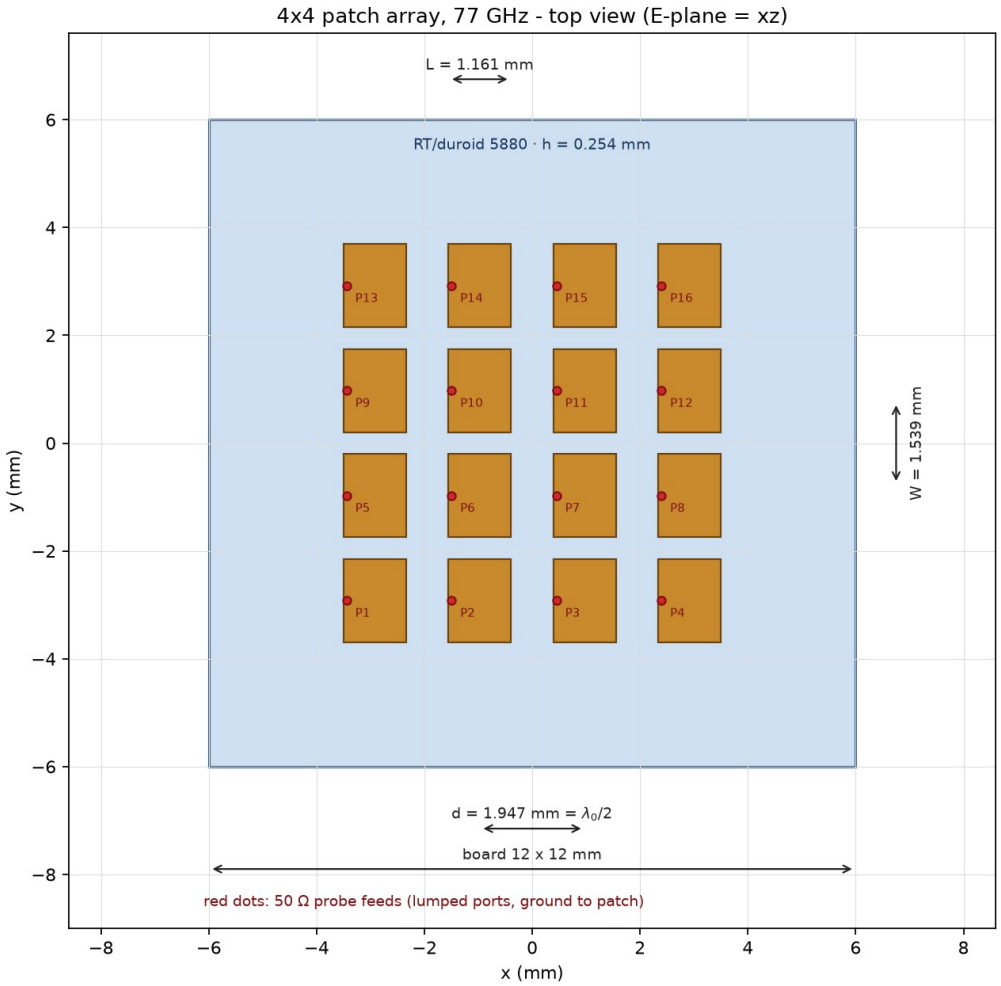


Figure 3. Top view of the 4×4 array with grid, patch and board dimensions. Red dots: 50 Ω probe ports.

3.4 Three-dimensional model

Figure 4 shows the complete simulation model. The patches sit on top of the laminate, the ground plane forms the bottom side and the probes pass through the substrate. The model is exactly what `array_model.py` writes into the openEMS script – there is no second, manually maintained geometry.

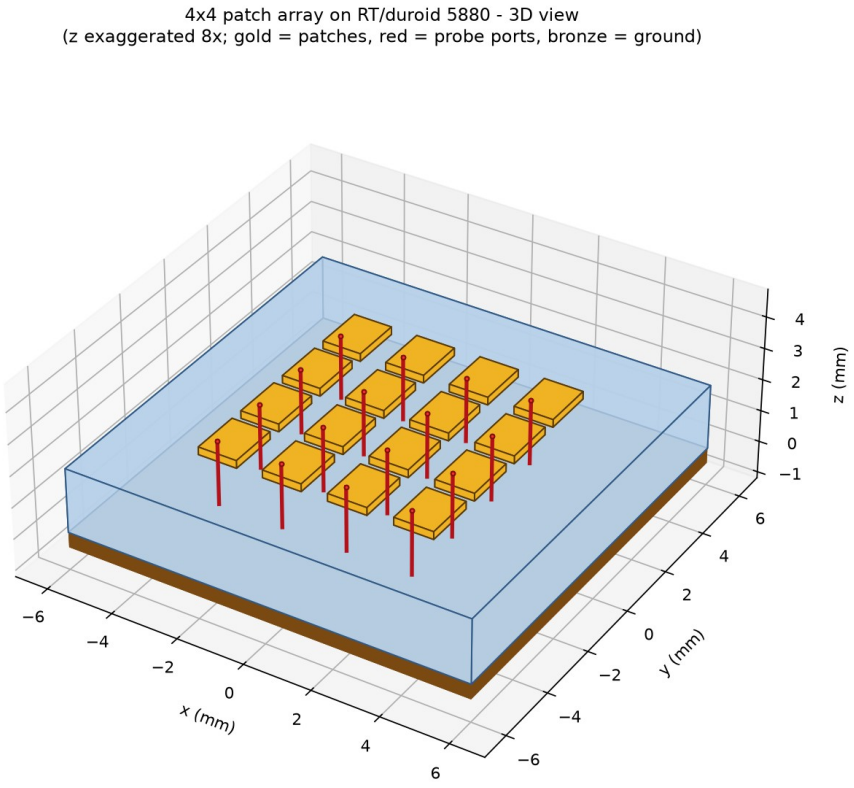


Figure 4. 3D model of the 4×4 array (z exaggerated 8×).

4 Element design and matching

4.1 First cut after Hammerstad

The Hammerstad design with L = 1.1383 mm and an inset of 30 % of the patch length (0.341 mm) is simulated as a reference. The result is sobering: at 77 GHz the input impedance is Zin = 14.8 + 68.0j Ω, a strongly inductive mismatch, and the reflection is only -1.8 dB. The large inset has transformed the impedance beyond the target.

It is notable that the reflection stays flat across the whole band – an indication that this is not a single mismatch but a systematically wrong feed impedance. Exactly this suspicion is tested with a control experiment in the next step.

4.2 Control experiment: open feed line

In the control run `probe_only` the feed is simulated without the patch: a short piece of coaxial line through the substrate, open at the top. An open line end is capacitive, a short-circuited one is inductive – the measurement therefore decides which equivalent circuit applies to the probe.

The left panel of Figure 5 shows the result: the measured reactance is strongly capacitive over the whole band (-205j Ω at 77 GHz) and follows a tangent curve. The fit of an open line yields Zp = 188 Ω, εp = 3.40 and an electrical length of 42.6° at 77 GHz; the deviation from the model is below 0.5 %.

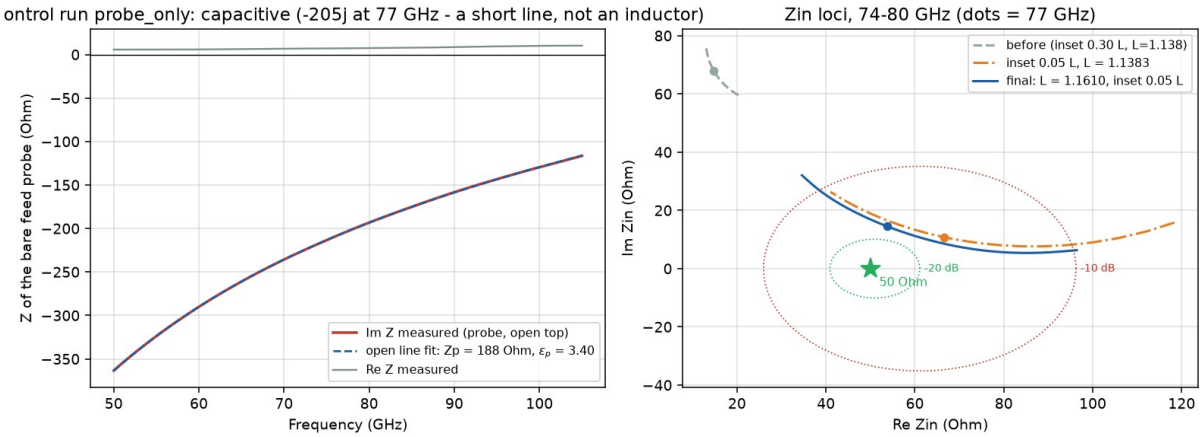


Figure 5. Control experiment and re-tuning: left the open feed line with its fit, right the input-impedance loci of the three design stages in the Z plane (dots: 77 GHz).

4.3 De-embedding and Newton re-tuning

With the line model the influence of the probe can be removed from the measured input impedance. What remains is the actual patch impedance, which can now be brought to 50 Ω in a targeted way via the two design parameters – patch length L (resonance) and inset (resistance). A local Newton iteration is applied to the simulated impedance map; the Jacobian comes from finite differences of the control runs already available.

Table 2 shows the iteration chain. Reducing the inset from 30 % to 5 % largely removes the inductive excess; the remaining deviation is corrected by lengthening the patch to 1.1610 mm. Two independent candidates were fully simulated as a check; candidate A is adopted as the final design.

Table 2. Design chain with simulated impedances at 77 GHz.

Step	L (mm)	Inset	Zin(77) (Ω)	|S11|(77) (dB)
Hammerstad first cut	1.1383	0.30 L	14.8 + 68.0j	-1.8
Inset control	1.1383	0.05 L	66.6 + 10.8j	-15.4
Candidate B	1.1683	0.02 L	38.7 + 20.7j	-11.7
Candidate A – final	1.1610	0.05 L	53.8 + 14.5j	-16.9

All values from the respective production or verification run, S-parameters at 77 GHz.

4.4 Final matching check

Figure 6 compares the final design with the first cut. The input impedance at 77 GHz is 53.8 + 14.5j Ω, the reflection -16.9 dB and the VSWR 1.33. The −10 dB bandwidth extends from 74.06 to 79.10 GHz, i.e. 5.04 GHz. The optimum is -17.4 dB at 76.58 GHz.

The comparison also shows the limit of the method: near the optimum the impedance map is almost collinear, i.e. length and inset move the impedance in the same direction. More than about −17 dB cannot be achieved with these two knobs at this substrate thickness; a deeper match would require an additional matching network or a coupled feed.

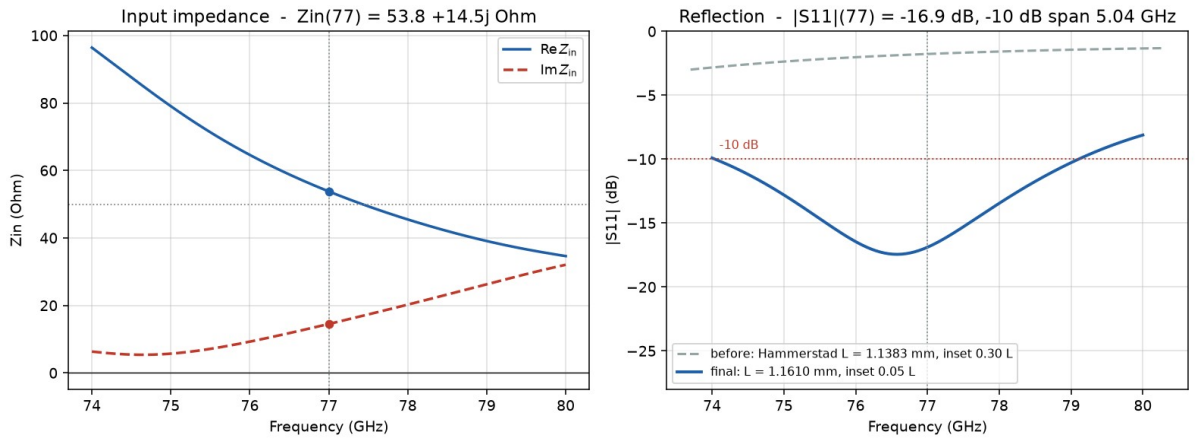


Figure 6. Input impedance (left) and reflection (right) of the final element; dashed: the Hammerstad first cut.

4.5 Pattern of the isolated element

The isolated element radiates with a directivity of 9.38 dBi at an efficiency of 0.939. Figure 7 shows the three-dimensional pattern; the beam points in +z as expected and the E plane is the xz plane.

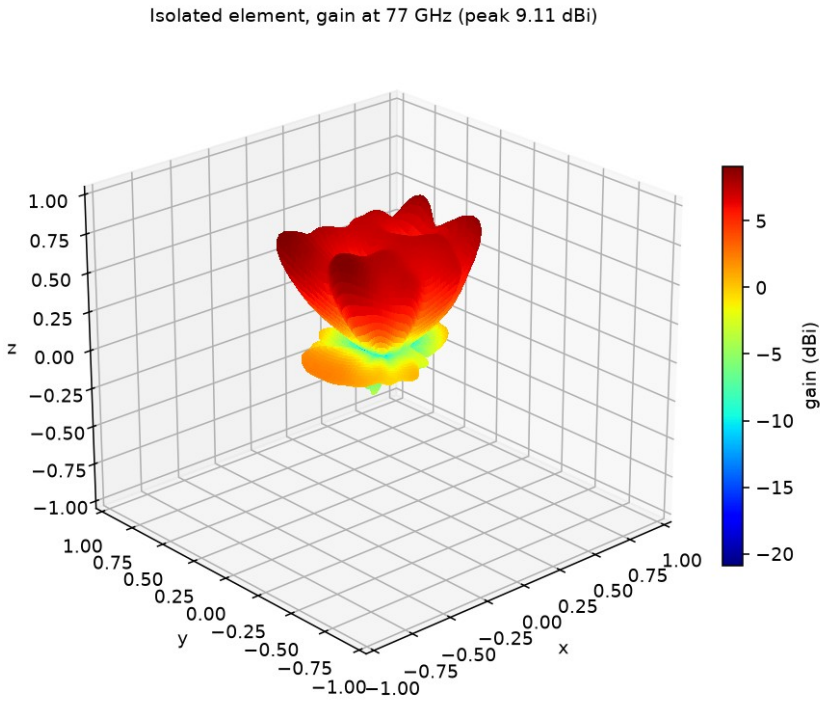


Figure 7. Pattern of the isolated element at 77 GHz (peak 9.11 dBi).

5 The 4×4 array

5.1 Embedding and coupling

In the `embedded` run only port 6 is excited while all other ports are passively terminated in 50 Ω. The run therefore yields the complete coupling column S_i6 and the embedded element pattern – the behaviour of one element in its real environment.

The self-reflection of the element is -18.0 dB, worse than that of the isolated element (-16.9 dB) because of neighbour coupling. The strongest coupler is the direct neighbour in x at -14.3 dB and 1.95 mm distance; the diagonal neighbours are around −25 dB and the elements at the board edge −28 dB and below. Figure 8 shows the coupling column versus distance and versus frequency.

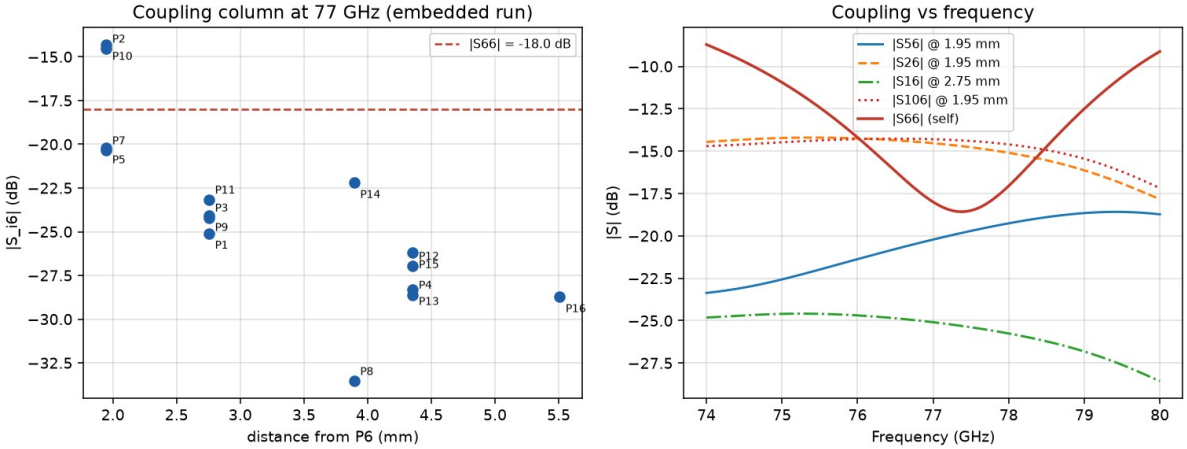


Figure 8. Coupling in the embedded run: column S_i6 at 77 GHz versus distance (left) and versus frequency (right).

The embedded element reaches a directivity of 9.16 dBi at an efficiency of 0.814. Figure 9 compares the E and H planes with the isolated element: the beam is slightly widened by the neighbours and the back lobe rises – the footprint of coupling, which is deliberately not modelled in the array-factor comparison later.

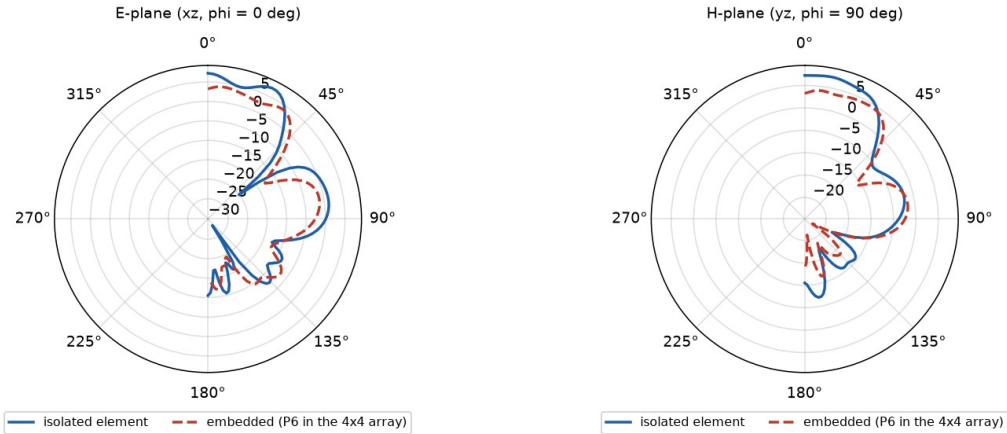


Figure 9. E and H plane at 77 GHz: isolated element (blue) versus embedded element P6 (red).

5.2 Broadside operation

In the `broadside` run all 16 ports are driven in phase. The directivity rises to 17.91 dBi. This agrees well with the aperture estimate $D \approx 4\pi A/\lambda_0^2 = 17.0$ dBi for a uniformly excited $\lambda_0/2$ aperture of $2\lambda_0 \times 2\lambda_0$; the 0.9 dB higher value follows from the element pattern, which tapers the edge of the visible range. The angular grid is 2°, and the maximum is at $\theta = 2°$, $\varphi = 0°$.

Figure 10 compares the full-wave pattern with the product of element pattern and array factor. The main beam agrees excellently down to −20 dB; the deviations in the nulls and the wide sidelobes are the signature of neglecting coupling and edge diffraction in the array-factor model.

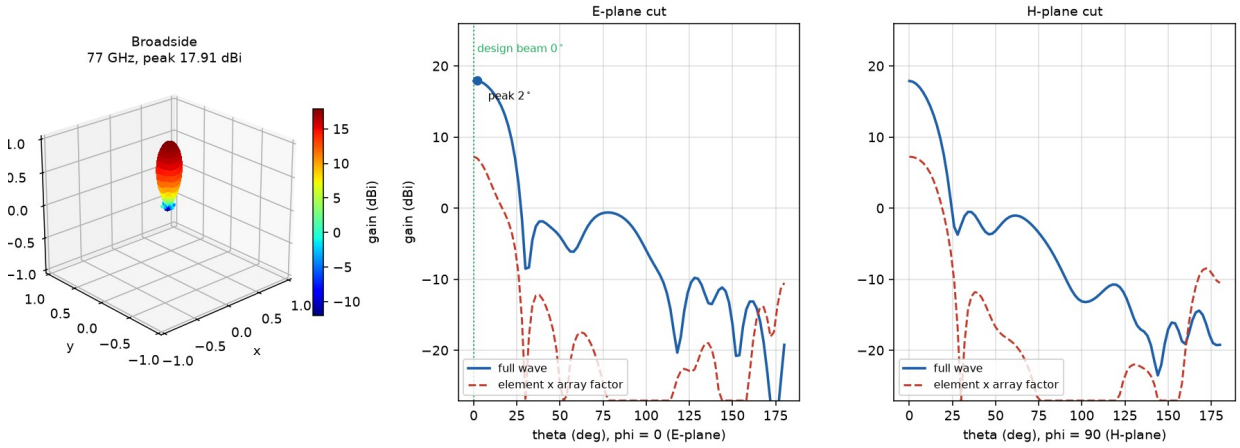


Figure 10. Broadside operation at 77 GHz: 3D pattern, E plane and H plane compared with element × array factor.

5.3 Beam steering by +30°

For the steered run each element in x receives a progressive phase of −90°, corresponding to 90.0° per spacing $d = \lambda_0/2$. The sign follows the openEMS phase convention: a wave travelling in +x carries $\exp(-jkx)$, so the excitation must lead. This convention was verified with a test run – with the opposite sign the beam appeared mirrored at $\varphi = 180°$.

The simulation confirms the steering: the maximum is at $\theta = 28°$, $\varphi = 0°$ with 17.75 dBi. The offset from the nominal 30° results from the finite element beam and mutual coupling; the array factor alone predicts 30°.

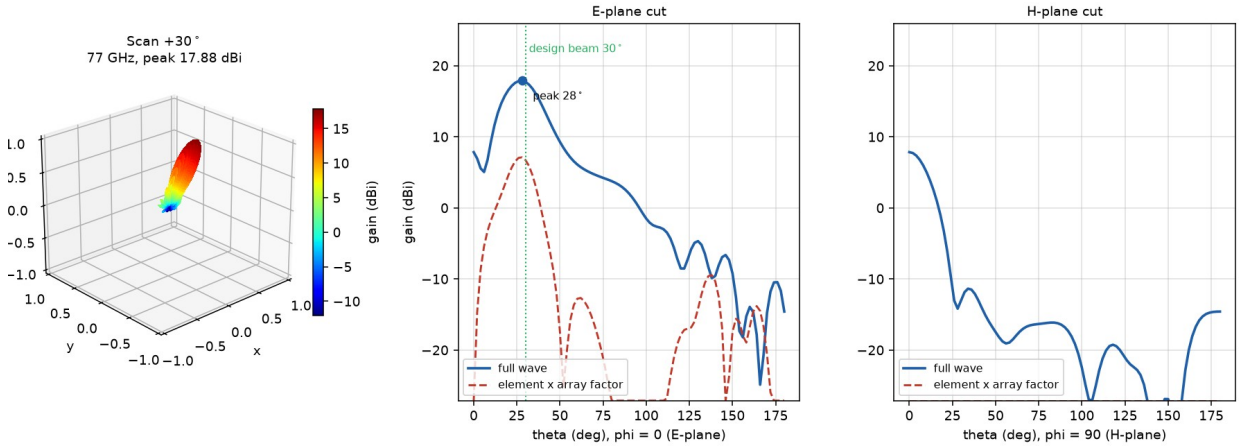


Figure 11. Steered operation (+30°): 3D pattern, E plane and H plane compared with element × array factor.

5.4 Comparison of both modes

Figure 12 overlays the E planes of both runs and additionally shows the active reflection of each port. The active reflection is the superposition of all returning waves at a port and therefore the relevant quantity in operation. In the broadside case it lies between −10.8 dB (ports 1 and 13) and −4.2 dB (ports 4, 8, 12 and 16); in the steered case it improves by 3 to 6 dB for most ports because the progressive phase partially cancels the coupling contributions. The right-hand column at x = +2.4 mm remains conspicuous at about −4 dB in both cases – there the neighbour couplings add up in phase.

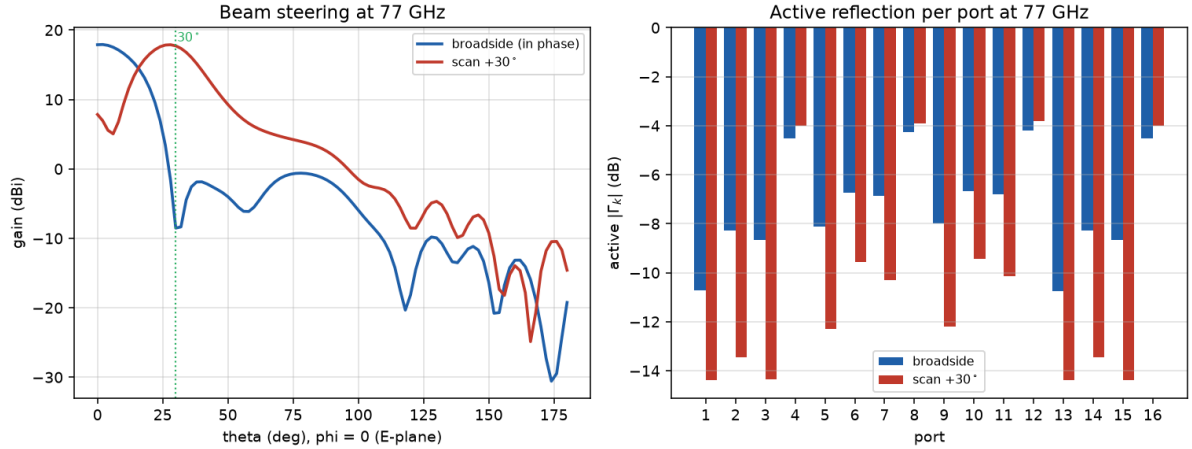


Figure 12. Broadside / +30° comparison: E plane (left) and active reflection per port at 77 GHz (right).

5.5 Near field

Figures 13 and 14 show snapshots of the electric field magnitude in broadside operation at 77 GHz, taken from the frequency-domain dumps of the run. In the E plane (Figure 13) the in-phase excitation of the four elements along x appears as a common wave front across the aperture; the field decays quickly with distance from the board, which limits near-field coupling between neighbouring elements. The top view at 1.02 mm height (Figure 14) shows the 4×4 aperture as a compact, contiguous source area – the patches still act as one radiating entity rather than 16 separate radiators.

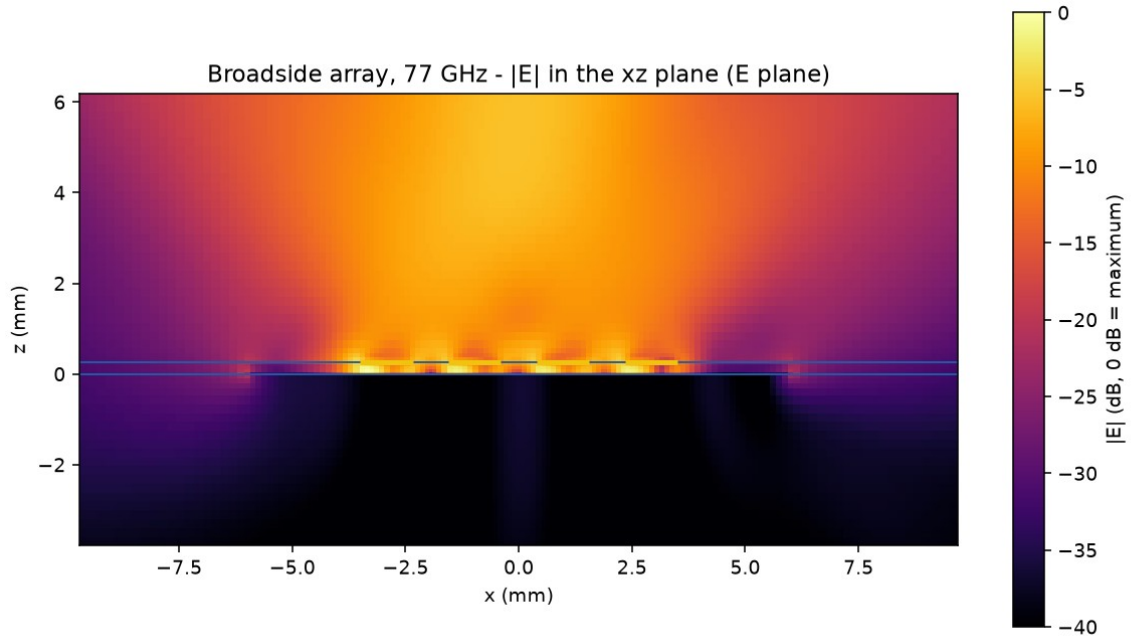


Figure 13. Electric field magnitude in the E plane (xz, y = 0) at 77 GHz, broadside operation. Yellow: patch edges.

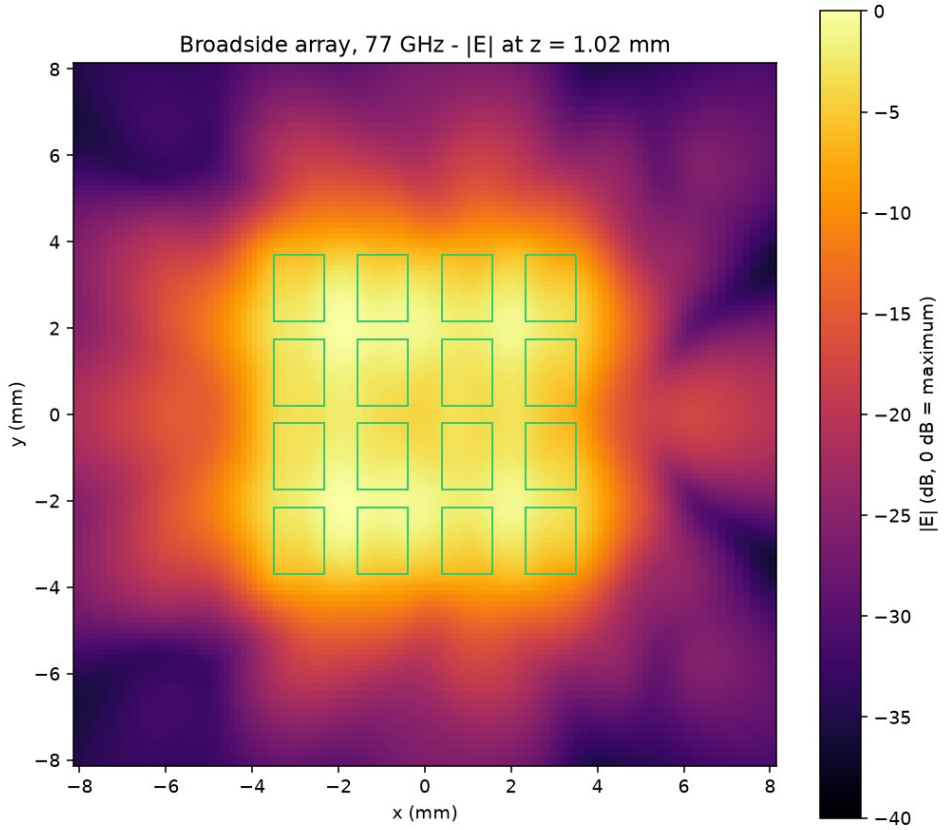


Figure 14. Top view of the field magnitude 1.02 mm above the board; green: patch outlines.

5.6 Beam-steering sweep

The animation is based on 25 complete FDTD runs with steering angles from −60° to +60° in 5° steps; only the progressive port phases change, the mesh and the board stay identical. Figure 15 shows the E planes of all runs on a signed angle axis (φ = 0 for +x, φ = 180 for −x). The maximum follows the nominal angle but lags increasingly behind it: about 1–2° at ±10°, 2–8° at ±30° and 18–26° at ±50…60°. The reason is the finite element beam, which rolls off towards large angles and pulls the product of element and array factor back towards broadside; the asymmetry between +x and −x follows from the slightly asymmetric pattern of the probe-fed patches. The peak gain stays at 17.6–18.2 dBi up to ±30° and drops towards the edges to 14.5 dBi (+60°) and 15.3 dBi (−60°). The θ0 = 0° run reproduces the broadside production run exactly (17.914 dBi at θ = 2°), which validates the sweep chain. The animation is available as `anim_beam_steer.webp`, `.gif` and `.mp4` in the repository.

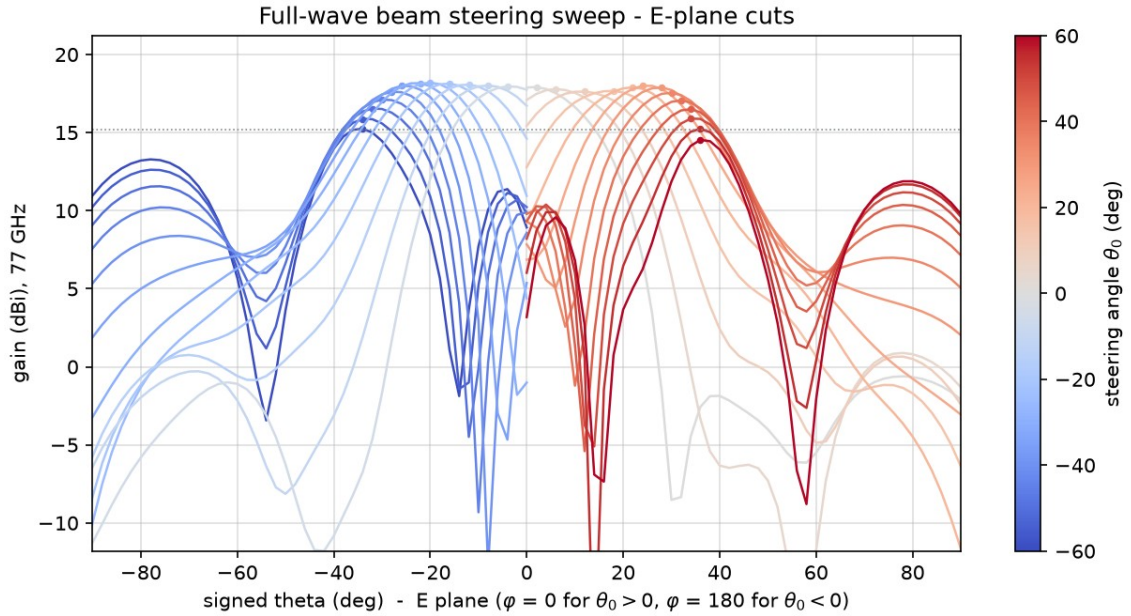


Figure 15. E planes of all 25 sweep runs (−60°…+60°) versus the signed angle (φ = 0 for +x, φ = 180 for −x); colour = nominal angle, dots = individual maxima.

6 Results and discussion

6.1 Key figures

Table 3. Summary of the array figures at 77 GHz.

Quantity	Broadside	Scan +30°
Directivity Dmax	17.91 dBi	17.88 dBi
Efficiency η (raw, 16-port artefact)	13.457	13.123
Efficiency η (corrected)	0.949	0.970
Gain (corrected)	17.69 dBi	17.75 dBi
Beam direction	θ = 2°, φ = 0°	θ = 28°, φ = 0°

Corrected η includes the accepted power of all 16 ports (see Section 6.3).

6.2 Accuracy and limitations

- The mesh resolves the 0.254 mm laminate with four cells and the metal edges with about three cells; the remaining numerical uncertainty of the directivity is about ±0.2 dB.
- The array-factor comparison ignores coupling and edge diffraction. The good agreement of the main beam shows that these effects shape the pattern but do not dominate the beam direction.
- Probe matching with two design knobs is limited to about −17 dB; a deeper match would require a matching network.
- Active reflection reaches only about −4 dB at the right-hand array edge; a real module needs feed matching or an edge treatment there.

6.3 Note on the power budget

The post-processing shipped with openEMS determines the accepted power from the port with the largest amplitude. This is correct for a single driven port, but for 16 simultaneously driven ports it underestimates the power by a factor of 16 and inflates efficiency and gain by about 12 dB. The corrected values quoted in this report sum the accepted power over all 16 ports; directivity and patterns are unaffected because they are pure field integrals.

When evaluating the port time series it must also be noted that openEMS stores voltage and current on slightly offset time grids. Transforming each signal on its own time axis introduces an apparent phase shift and hence an error in the real power of up to 11 %. The evaluation therefore uses the same time axis for both signals, exactly as the openEMS post-processing does.

7 Reproduction

7.1 Prerequisites

Python 3.11 with numpy, matplotlib and h5py is required, plus an openEMS installation reachable through the VisualopenEMS project layer. The path to the VisualopenEMS root is set via the environment variable `VISUALOPENEMS_ROOT`:

```
VISUALOPENEMS_ROOT = C:\path\to\FreeCAD_OpenEMS_GUI
```

7.2 Workflow

```
python make_project.py --preview # projects + figures 01..04
python validate_project.py # mesh / collision checks
python run_headless.py patch_element_77ghz.fcsim.json
python run_headless.py patch_array_4x4_77ghz.fcsim.json
python run_headless.py patch_array_4x4_broadside.fcsim.json
python run_headless.py patch_array_4x4_scan30.fcsim.json
```

```
python plot_results.py # figures 05..14 + results.json
python build_docs.py # this report (DE + EN)
```

7.3 Computational effort

Table 4. Computational effort of the four production runs.

Run	Grid (cells)	Time steps	Wall time
Single element	154×155×75 = 1 790 250	10 506	approx. 20–30 min
Embedded element	163×165×75 = 2 017 125	10 598	approx. 20–30 min
Broadside array	163×165×75 = 2 017 125	10 598	approx. 20–30 min
Array +30°	163×165×75 = 2 017 125	10 598	approx. 20–30 min

Cell counts from the openEMS start-up message, time steps from the energy time series; wall time on an 8-core laptop.

7.4 Files

```
array_model.py single source of truth
make_project.py project writer + construction figures
validate_project.py pre-run checker
run_headless.py openEMS runner
plot_results.py result figures + results.json
build_docs.py this report
*.fcsim.json the four projects
work_*/ simulation output
docs/figures/ figure 01..14
docs/results.json numeric source for the report
```